

SOVETSKAYA GAVAN, Russian Federation

WMO: 317700

Lat: 49.000N

Lon: 140.300E

Elev: 78

StdP: 14.65

Time Zone: 10.00 (E10)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-10.6	-24.8	1.4	-11.6	-21.7	1.7	-8.2	31.5	15.7	27.4	16.2	4.0	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.2	78.8	66.0	75.0	64.6	71.6	62.9	68.1	76.0	66.0	72.5	64.3	69.6	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	93.3	71.4	63.5	87.9	69.2	62.1	83.5	67.2	32.6	76.6	30.9	72.6	29.6	69.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.2	22.4	18.3	DB	-20.3	86.7	3.7	2.5	-22.9	88.5	-25.1	89.9	-27.2	91.3	-29.9	93.2	(4)
(5)			WB	-20.6	72.5	3.6	1.6	-23.2	73.6	-25.4	74.6	-27.4	75.5	-30.1	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	5.3	8.4	21.2	34.3	44.7	52.9	60.7	63.7	56.5	42.7	24.6	10.3
	DBStd	21.42	8.20	8.50	7.63	6.12	5.95	5.95	5.05	4.21	5.99	6.37	9.92	8.63
	HDD50	6371	1385	1166	892	471	185	32	0	0	13	236	761	1230
	HDD65	10791	1850	1586	1357	920	629	365	150	76	261	691	1211	1695
	CDD50	1116	0	0	0	1	21	119	333	424	208	10	0	0
	CDD65	62	0	0	0	0	0	3	18	35	6	0	0	0
	CDH74	411	0	0	0	0	7	52	130	172	50	0	0	0
	CDH80	57	0	0	0	0	0	7	26	19	5	0	0	0
Wind	WSAvg	6.7	7.4	6.3	6.9	7.0	5.8	4.8	4.6	5.0	6.0	8.1	8.8	9.2
Precipitation	PrecAvg	29.1	0.6	0.8	1.6	2.1	2.8	2.8	3.0	4.2	4.5	3.7	1.8	1.4
	PrecMax	50.9	1.7	3.2	4.6	6.2	5.4	7.6	9.1	21.1	13.3	8.9	5.4	6.5
	PrecMin	21.5	0.0	0.0	0.2	0.4	0.6	0.3	0.1	0.5	0.0	0.2	0.0	0.0
	PrecStd	6.5	0.5	0.8	1.0	1.2	1.3	1.7	2.0	3.4	2.8	2.2	1.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	34.3	45.6	64.7	73.0	80.8	83.6	82.7	79.7	65.2	50.9	33.1
		MCWB	29.3	27.4	36.2	46.8	53.9	64.1	70.1	68.8	65.3	51.5	41.4	31.5
	2%	DB	25.8	28.5	39.7	54.8	65.9	73.4	77.5	78.5	74.1	60.3	45.4	30.0
		MCWB	23.3	24.5	32.7	42.6	51.9	61.3	66.9	66.7	62.5	50.1	38.1	27.9
	5%	DB	22.4	25.1	35.8	48.4	60.7	68.5	72.7	75.2	70.0	56.0	41.4	27.0
		MCWB	19.7	22.1	30.8	39.1	48.7	59.0	64.8	65.7	60.4	48.6	37.3	24.7
	10%	DB	18.8	22.2	32.9	43.9	55.6	63.9	69.1	71.8	66.2	53.0	38.8	24.2
		MCWB	16.2	19.6	29.1	37.0	47.0	56.2	62.9	64.5	58.5	46.4	34.6	21.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.1	28.5	36.7	47.7	56.7	66.4	71.0	71.1	66.6	56.1	42.4	32.1
		MCDB	30.0	32.6	43.7	61.8	70.2	77.2	82.4	78.1	76.4	59.8	47.9	32.8
	2%	WB	23.3	25.1	33.6	43.9	52.6	62.5	67.9	68.8	64.2	52.2	39.8	28.2
		MCDB	25.5	28.0	37.6	53.1	63.0	72.8	76.3	75.7	70.8	57.1	43.8	29.7
	5%	WB	19.8	22.5	31.5	39.8	49.9	59.5	65.4	66.7	62.3	50.0	37.7	25.1
		MCDB	22.3	25.0	35.1	46.9	58.2	66.9	71.4	72.8	67.2	54.0	41.2	27.1
	10%	WB	16.2	19.5	29.5	37.1	47.7	56.8	63.4	65.1	60.3	47.6	35.1	21.8
		MCDB	18.8	22.3	32.7	42.7	54.6	62.9	68.2	70.3	64.6	52.4	39.2	24.0
Mean Daily Temperature Range	5% DB	MDBR	14.9	18.3	16.4	13.8	15.2	14.7	13.5	13.2	16.2	15.3	13.5	13.9
		MCDBR	12.3	19.7	18.6	24.4	27.2	25.4	22.5	20.9	21.9	21.6	16.0	13.7
		MCWBR	11.0	16.3	13.5	14.0	14.2	14.0	13.0	10.7	11.9	13.8	12.0	12.8
	5% WB	MCDBR	10.1	17.5	17.0	22.9	23.2	23.2	20.7	17.8	17.8	18.4	14.4	12.6
		MCWBR	9.4	14.8	12.9	13.6	13.3	13.6	12.5	10.2	10.8	13.2	11.5	12.3
Clear-Sky Solar Irradiance	taub	0.289	0.323	0.394	0.486	0.532	0.505	0.473	0.428	0.387	0.393	0.347	0.301	
	taud	2.145	2.031	1.902	1.828	1.845	2.022	2.178	2.324	2.365	2.165	2.147	2.134	
	Ebn at Noon	236	253	251	240	235	244	250	257	257	226	208	211	
	Edh at Noon	26	37	50	61	62	53	44	37	32	33	26	23	
All-Sky Solar Radiation	RadAvg	429	779	1165	1397	1520	1615	1479	1349	1155	753	441	321	
	RadStd	45	60	93	155	123	172	147	125	109	49	36	36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	+1.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page